



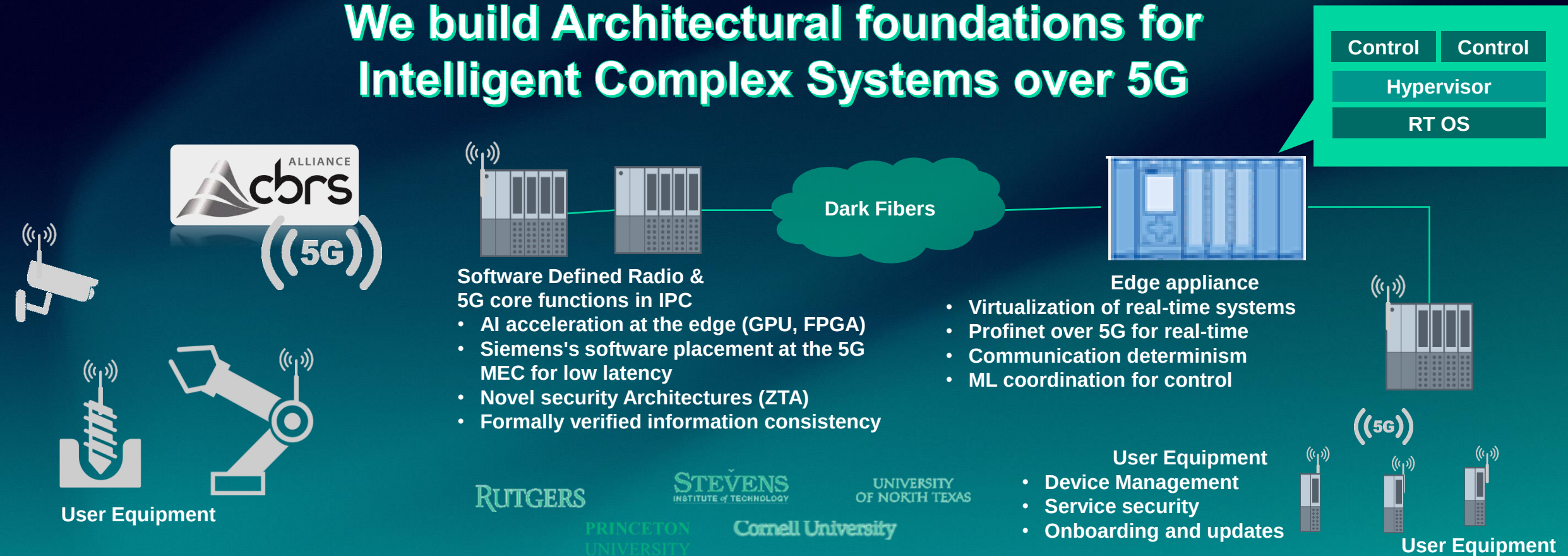
NDN for Next Generation of Factory Automation

NDNComm meeting 2021

By Dr. Charif Mahmoudi

Emergence of Intelligence from the Complexity of the Factory Floor

We build Architectural foundations for Intelligent Complex Systems over 5G



Fostering architectures for deterministic communicating Intelligent Systems

NDN as Paradigm for Factory Automation ?

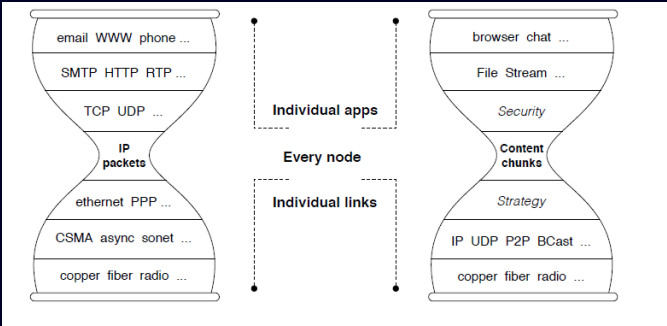
Hard Real-Time Requirements



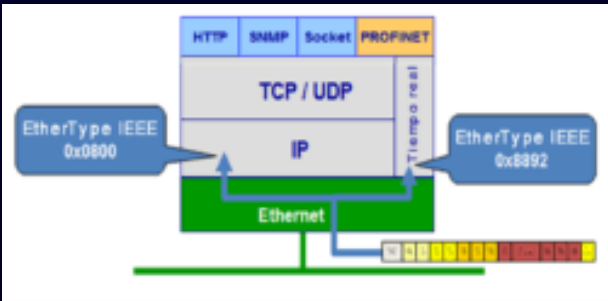
Protocol support



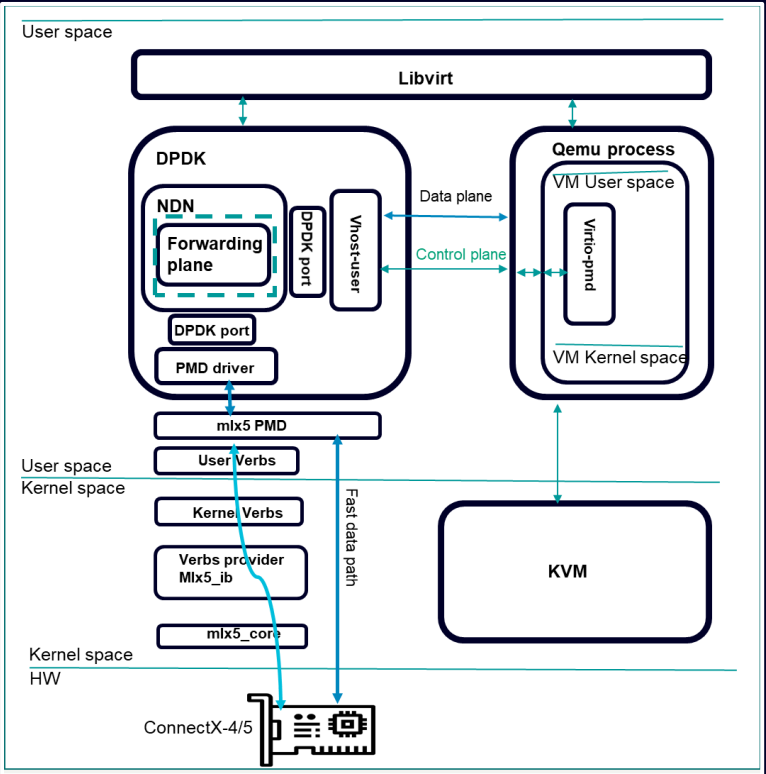
Model Parallelism to IP



Real-Time Feature



<https://us.profinet.com/is-profinet-deterministic/>



Ayadi, Mohamed Issam, et al. "Deep Learning Forwarding in NDN With a Case Study of Ethernet LAN." IJWLTT 16.1 (2021): 1-9.
Khoussi, Siham, et al. "Performance evaluation of the ndn data plane using statistical model checking Springer, Cham, 2019.
Dongo, Junior, Charif Mahmoudi, and Fabrice Mourlin. "Ndn log analysis using big data techniques: Nfd performance assessment." IEEE, 2018.

| Contact

Published by Siemens Technology

Charif Mahmoudi

Software Architect for Intelligent Systems

Architecture, Verification, and Intelligent Systems Group

Siemens Technology

Princeton, NJ

USA

E-mail charif.mahmoudi@siemens.com